

Complain / Ticket Management System

Architecture & Functional Overview

Document Status: Draft for Development Review

Platform: Oracle APEX / Oracle Database

Purpose: Functional and architectural definition before development

1. Purpose

This document describes the intended structure and workflow of a **Complain / Ticket Management System** designed to manage complaints, service requests, operational issues, assignments, progress, closure and user onboarding across multiple organizations and departments.

The system will provide controlled access through three primary categories:

- **Super Admin**
- **Organization-wise Admins**
- **Organization-wise Users**

The system will maintain a complete activity trail for ticket creation, assignment, status changes, comments, attachments, closure requests, approvals and other important actions.

The objective is to establish the complete business flow and access model before finalizing the database structure and starting development.

2. User & Access Hierarchy

The system will follow an organization-based access model.

2.1 Super Admin

The Super Admin has system-wide visibility and control.

Responsibilities

- View all organizations.
- Manage organizations.
- Create/manage Organization Admins.
- View users across all organizations.
- Approve or reject organizational user registration requests.

- View all tickets/complaints.
- Manage global ticket types.
- Manage priorities and common statuses.
- Monitor ticket activity.
- Reassign or override tickets when required.
- Access system-wide reports and dashboards.
- Manage departmental configuration.
- Manage registration links/forms.
- Maintain system-level configuration.

The Super Admin represents the highest level of administrative control.

3. Organization-wise Admin

Each organization may have one or more administrators.

An Organization Admin is restricted to the organization to which the administrator belongs.

Responsibilities

- View tickets belonging to their organization.
- Create complaints/tickets.
- Assign tickets to users within their organization.
- Assign one ticket to multiple users where required.
- Monitor assigned users and ticket progress.
- Review closure requests.
- Reopen or mark a ticket as **Not Completed** where appropriate.
- View organization users.
- Review organizational registration requests where permitted.
- Monitor ticket activity.
- View organization-level reports and dashboards.
- Manage organization-specific ticket types, departments, priorities or categories where authorized.

An Organization Admin **cannot access another organization's operational data** unless explicitly granted Super Admin privileges.

4. Organization-wise Users

Regular users belong to a specific organization and department.

Responsibilities

- View tickets assigned to them.

- Update the progress/status of tickets assigned to them.
- Add remarks/comments.
- Upload supporting attachments.
- Raise a new complaint/ticket where permitted.
- View tickets personally raised by them.
- Submit a closure request when their work is completed.

A regular user cannot directly close a ticket.

The final closure remains subject to the defined approval workflow.

5. Organization & Department Structure

The system will maintain an organizational hierarchy.

Proposed hierarchy

Organization

↓

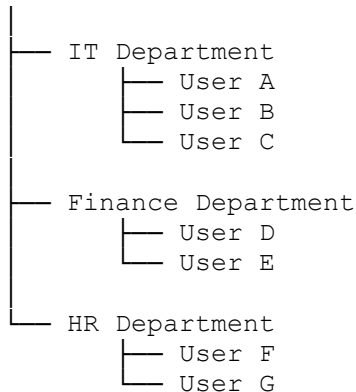
Department

↓

User

For example:

Organization



This hierarchy will be important for both **ticket assignment** and **user registration**.

6. Department-wise User Registration

New user registration will not be a generic unrestricted registration process.

The registration form will be associated with an **organization and department**.

There will be two possible mechanisms.

6.1 Department-specific Registration Link

A registration link may be generated or distributed for a particular department.

For example:

```
Organization: ABC
Department: IT
Registration Link
    ↓
Registration Form
    ↓
Department = IT
```

The applicant will therefore be registered against the appropriate organization and department.

6.2 Organization Registration Link

Alternatively, an organization-level registration link may be distributed.

When the applicant opens the registration form, the form will display/select the applicable department.

For example:

```
Organization
    ↓
Department
    ↓
User Information
    ↓
Submit Registration
```

The registration request must therefore always contain the organizational and departmental context.

7. New User Registration Workflow

The registration workflow will be:

```
Registration Link
    ↓
```

Registration Form
↓
Organization / Department
↓
Applicant Details
↓
Submit
↓
Pending Registration
↓
Internal Notification
↓
Registration Request Logged
↓
Admin Review
↓
Approve / Reject

When a new organizational user submits the registration form:

1. The registration request is stored in the system.
2. The request is placed in **Pending** status.
3. The responsible internal recipient receives an email notification that a new registration request has been received.
4. The registration request is displayed in the administrative dashboard.
5. The request can be reviewed and processed by the authorized administrator.
6. The registration request remains fully logged for audit purposes.

8. Email Notification Policy

Email notifications should be kept deliberately simple.

The system will **not send an email for every approval, rejection or ticket status change**.

Routine status changes will be visible through:

- Dashboard
- Notification area/inbox
- Ticket activity
- User's pending/action queues
- Registration management screen

8.1 Email-triggering Event

The primary email-triggering event will be:

New User Registration

When a new user registers through an organizational/departmental registration form:

New Registration
↓
Registration Stored
↓
Email to Responsible Internal Recipient
↓
Registration Request Appears in Dashboard

The email should contain only essential information, such as:

- Organization
- Department
- Applicant name
- Applicant email
- Registration date/time
- Registration reference/request number
- Action required: Review registration request

The email should **not itself approve the user**.

The actual approval/rejection remains an application-level action.

9. Ticket / Complaint Management

A ticket represents a complaint, issue, request or operational task that requires action and tracking.

A ticket may be created by:

- Super Admin
- Organization Admin
- Authorized Organization User

The ticket should contain, where applicable:

- Ticket/Complaint Number
- Organization
- Department
- Customer / Related Entity
- Ticket Type
- Category
- Priority
- Description
- Created By
- Assigned User(s)
- Creation Date/Time
- Current Status
- Attachments

- Remarks
 - Activity History
-

10. Ticket Assignment

An administrator may assign a ticket to:

- One user
- Multiple users

For multiple users, each user will have an individual responsibility/status against the same ticket.

Example:

```
Ticket #10025
├── User A → Completed
├── User B → In Progress
└── User C → Pending
```

The system will maintain the individual progress of each assigned user while also maintaining an overall ticket status.

11. User-raised Complaints

An authorized organization user may raise a complaint directly.

The user does not need an administrator to create the ticket first.

The system will record:

- Who created the ticket
- Organization
- Department
- Date/time
- Ticket category
- Priority
- Description
- Attachments
- Assigned user(s)

Administrators will retain visibility of tickets raised by their organization users.

12. Ticket Status Workflow

A proposed status flow is:

```
New
↓
Assigned
↓
In Progress
↓
Pending Closure
↓
Closed
```

Where necessary:

```
Pending Closure
↓
Admin Review
↙   ↘
Closed Reopened / Not Completed
      ↓
      In Progress
```

The exact status list can be finalized during the database/status workflow design.

13. Closure Request

A regular user will not directly close a ticket.

When the assigned work is completed:

```
User completes work
↓
User submits Closure Request
↓
Ticket = Pending Closure
↓
Admin Review
↓
Approve → Closed
Reject → Reopened / Not Completed
```

The closure request and the administrator's final decision will be recorded in the activity history.

14. Reopen / Not Completed

If an administrator reviews a closure request and determines that the work has not actually been completed, the administrator may mark the ticket:

Not Completed / Reopened

A remark should be mandatory when performing this action.

Example:

```
Closure Requested
      ↓
Admin Review
      ↓
Not Completed
      ↓
Admin Remark
      ↓
Assigned User
      ↓
Continue Work
```

This action must be recorded with:

- User performing the action
- Date/time
- Previous status
- New status
- Remark

15. Activity & Audit Logging

The system should maintain a complete activity history.

The following actions should be logged:

- Ticket creation
- Ticket assignment
- Assignment changes
- Status changes
- Remarks/comments
- Attachments
- Closure request
- Closure approval
- Closure rejection
- Reopen / Not Completed
- User registration

- Registration approval
- Registration rejection
- Other administrative actions

Each activity should contain, where applicable:

- Action
- Performed By
- Date/Time
- Previous Value
- New Value
- Remarks
- Reference to the relevant ticket/request

The activity log is an important accountability and audit mechanism.

16. Attachments

Attachments may be added during:

- Ticket creation
- Ticket update
- User response
- Closure request

Each attachment should record:

- File name
- File type
- File size
- Uploaded by
- Upload date/time
- Related ticket/request

Allowed file types and maximum file size should be finalized before implementation.

17. Dashboards

17.1 Super Admin Dashboard

The Super Admin dashboard may provide:

- Total Organizations

- Total Organization Admins
 - Total Users
 - Pending Registrations
 - Total Tickets
 - Open Tickets
 - Pending Closure
 - Closed Tickets
 - Reopened Tickets
 - Organization-wise statistics
 - Department-wise statistics
 - Ticket priority statistics
-

17.2 Organization Admin Dashboard

The Organization Admin dashboard should focus only on the administrator's organization.

Possible information:

- Total tickets
 - New tickets
 - Assigned tickets
 - In-progress tickets
 - Pending closure
 - Closed tickets
 - Reopened tickets
 - Department-wise ticket distribution
 - User workload
 - Pending registration requests
 - Recent activity
-

17.3 User Dashboard

A regular user should see:

- Tickets assigned to me
- Tickets raised by me
- In-progress tickets
- Pending actions
- Closure requests
- Recently updated tickets

The user should not have visibility of unrelated organizational data.

18. Access Control

The system must enforce organization-level data security.

The fundamental rule is:

```
Super Admin
  ↓
All Organizations

Organization Admin
  ↓
Own Organization
  ↓
Departments / Users / Tickets

Organization User
  ↓
Permitted Tickets
  ↓
Own Raised Tickets
```

Application authorization and database-level controls should prevent unauthorized cross-organization access.

19. Notifications

The system will primarily use **in-application visibility** rather than sending an email for every event.

Application notifications / dashboard visibility

Examples:

- Ticket assigned
- Ticket status changed
- Closure requested
- Closure rejected
- Ticket reopened
- Administrative action

Email

Email will primarily be used for:

- New organizational user registration notification

This approach avoids excessive email traffic while keeping important operational information visible inside the system.

20. Registration & Ticket References

Each registration request and ticket should have a unique system-generated reference.

Example:

Registration:
REG-2026-000125

Ticket:
TKT-2026-001025

These references should be used in:

- Dashboard
 - Activity log
 - Search
 - Reports
 - Email notification
 - Administrative processing
-

21. Reporting

The system should eventually provide reports based on:

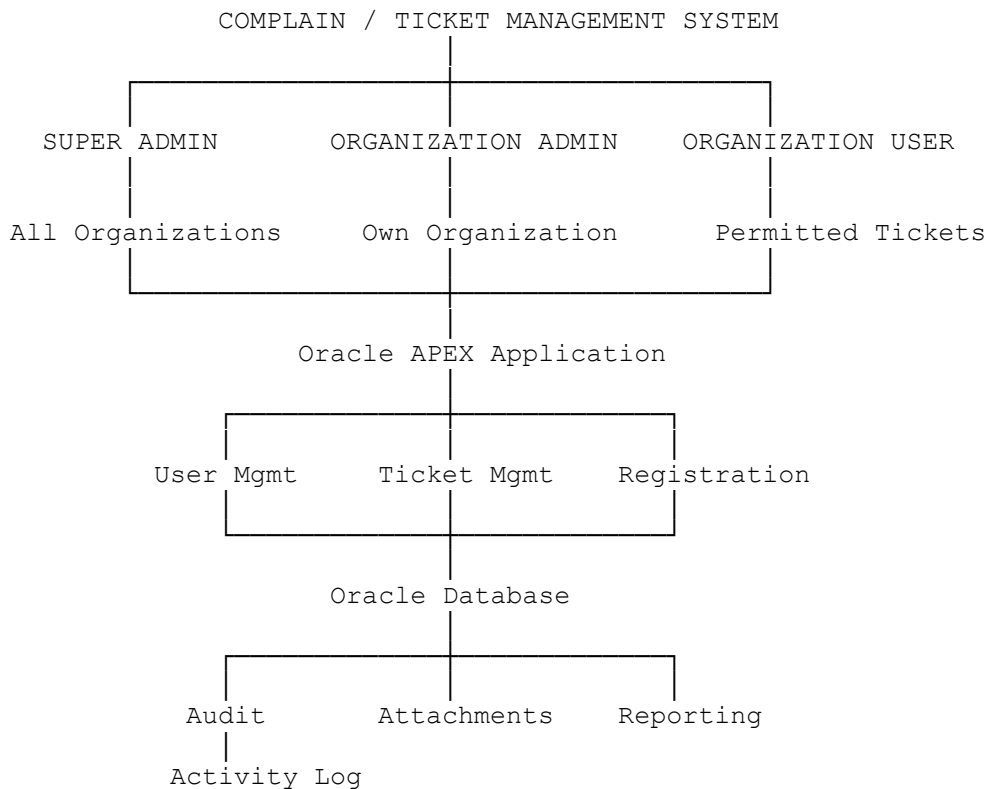
- Organization
- Department
- User
- Ticket type
- Category
- Priority
- Status
- Creation date
- Closure date
- Assigned user
- Registration status

Examples:

- Organization-wise ticket summary
- Department-wise complaint summary
- User workload report

- Open ticket aging report
- Closed ticket report
- Pending closure report
- Reopened ticket report
- Registration request report
- Ticket activity/audit report

22. Proposed High-Level Architecture



23. Key Business Principles

The system should be designed around the following principles:

1. Organization isolation

Users should only see information they are authorized to access.

2. Department awareness

Users and registration requests should belong to a defined organization and department.

3. Controlled closure

Users request closure; authorized administrators confirm it.

4. Complete auditability

Important actions must be recorded with user and timestamp.

5. Controlled notifications

Email should not be generated for every status change. Routine information should remain inside the application.

6. Registration accountability

Every new registration request should be recorded and traceable.

7. Multi-user assignment

A single ticket may require work from multiple users.

8. Administrative visibility

Administrators should have sufficient visibility to monitor workload, progress and unresolved complaints.